Toghrul Karimov

toghs@mpi-sws.org toghrul-karimov.github.io

Personal

Computer scientist working on algorithmic problems at the intersection of formal verification, dynamical systems, and number theory

Born: 09/1998

Citizenship: Azerbaijan, German naturalisation in progress

German language skills: C1

Academic employment

05/2025 – present

Postdoctoral researcher at the *Institut de Recherche en Informatique Fondamentale* (**IRIF**, Paris) and the Max Planck Institute for Software Systems (**MPI-SWS**, Saarbrücken)

Funded by the ERC Synergy Grant **DynAMiCs**

03/2024 – 04/2025

Postdoctoral researcher at the **MPI-SWS**

Funded by the German Research Foundation (DFG) through TRR 248 "Foundations of Perspicuous Software Systems" (**CPEC**, grant between TU Dresden and Saarland Informatics)

Education

2019 – 2024

PhD, **Saarland University** and the **MPI-SWS**, Saarbrücken

Supervisor: Joël Ouaknine

Thesis: Algorithmic verification of linear dynamical systems (doi)

Nominated for the *GI-Dissertationspreis* (one nomination from each university in Germany, Austria, and Switzerland across entire computer science), see below for awards

2015 – 2019

MCompSci Computer Science, **University of Oxford**

First Class Honours in both the bachelor's and the master's parts

Academic references

Joël Ouaknine (MPI-SWS), Valérie Berthé (IRIF), Joost-Pieter Katoen (RWTH Aachen), Laura Kovács (TU Wien), James Worrell (University of Oxford), Christel Baier (TU Dresden)

Awards and scholarships

Otto Hahn Medal of the Max Planck Society 2026

For fundamental contributions to automated formal verification of linear dynamical systems; awarded annually for outstanding doctoral research to up to 10 young scientists in the Chemistry, Physics and Technology Section of the Max Planck Society, 32 institutes in total

Ackermann Award 2025

EACSL Outstanding Dissertation Award for Logic in Computer Science

Fellowship of the Hausdorff Research Institute for Mathematics, Bonn 2025

For the duration of the Trimester Program "Definability, Decidability, and Computability"

ACM SIGBED Best Paper Award 2024

HSCC 2024, for the co-authored paper *Linear dynamical systems with continuous weight functions*

Distinguished Paper Award 2024

LICS 2024, for the co-authored paper *On the decidability of monadic second-order logic with arithmetic predicates*

CPEC mini-project grant 2022

Funding for a two-week research visit to the University of Oxford

Keble College scholarship 2016–2019

University of Oxford; awarded for top academic performance at the end of each year

Scholarship of the Ministry of Education of Azerbaijan 2015–2019

Full funding for studies at the University of Oxford

Professional service

Member of the program committee of CSL 2027

Regular reviewer for LICS, ICALP, MFCS, STOC, FOCS, and SODA

Supervision and teaching

As a postdoctoral researcher, I work closely with the doctoral students of the group "Foundations of Algorithmic Verification" at the MPI-SWS.

05/2026 – 07/2026

Supervised Daria Goliasova, research intern at the MPI-SWS

Project: developing a polynomial-time algorithm and an efficient implementation using an SMT solver for nondeterministic linear loops with two variables (tool) (preprint)

09/2026 – 12/2026

"Operating systems" (undergraduate course), lecturer

Azerbaijan University of Architecture and Construction, online

02/2026 – 05/2026

"Principles of cybersecurity" (undergraduate course), lecturer, also responsible for practicals and tutorials

Azerbaijan University of Architecture and Construction, online (lecture notes)

02/2026 – 05/2026

"Machine learning" (undergraduate course), lecturer, also responsible for practicals and tutorials

Azerbaijan University of Architecture and Construction, online (lecture notes)

10/2022 – 03/2023

"Topics in algorithmic dynamical systems theory" (graduate course), teaching assistant

Saarland University

04/2020 – 09/2020

"Automata and sequences" (graduate course), teaching assistant

Saarland University

Invited talks

2026

Linear dynamical systems under the lens of o-minimality

Symbolic Dynamics and Arithmetic Expansions, Roscoff, France (workshop)

Preservation theorems for transducer outputs

20th Mons Theoretical Computer Science Days, Napoli, Italy (conference)

Rich sequences and decidability of logical theories

Algebraic Aspects of Symbolic Dynamics, Porto, Portugal (conference)

Automata on S-adic words

One World Numeration Seminar, online (seminar)

Rich sequences and decidability of logical theories

Mathematical and Algorithmic Aspects of Dynamical Systems, Bellairs Research Institute, Barbados (workshop)

Algorithmic verification of linear dynamical systems

Computer Science Logic (CSL), Paris, France (presentation for the Ackermann Award; slides)

2025

Decidability of logical theories via rigidity and randomness in dynamical systems
Algebraic Methods in Dynamics and Particle Physics, Saarbrücken, Germany (slides)

Some applications of o-minimality to computational problems in dynamical systems theory
Hausdorff Research Institute for Mathematics, Bonn, Germany (video)

Arithmetic predicates and decidability of logical theories
Hausdorff Research Institute for Mathematics, Bonn, Germany (video)

Extensions of linear programming with squares and powers of integers
Kyoto University, Kyoto, Japan

From word combinatorics to automatic structures
Workshop on Recent Developments in Arithmetic Theories and Applications, online (workshop)

Ode to o-minimality
Stellenbosch Logic Seminar, online (slides)

2023

The model-checking problem for linear dynamical systems
Workshop on Algorithmic Aspects of Dynamical Systems, Bellairs Research Institute, Barbados (workshop)

Conference and journal papers

The authors are, in all cases, listed alphabetically. The papers for which I was the clear lead author are marked with an asterisk.

*Automata on S-adic words (doi)
V. Berthé, T. Karimov, and M. Vahanwala
ICALP 2026

*Linear dynamical systems with weight functions (doi)
R. Aghamov, C. Baier, T. Karimov, J. Ouaknine, and J. Piribauer
Nonlinear Analysis: Hybrid Systems, Vol. 60 (2026)

Model checking linear temporal logic with standpoint modalities (doi)
R. Aghamov, C. Baier, T. Karimov, R. Majumdar, J. Ouaknine, J. Piribauer, and T. Spork
KR 2025

Multiple reachability in linear dynamical systems (doi)
T. Karimov, E. Kelmendi, J. Ouaknine, and J. Worrell
LICS 2025

*Verification of linear dynamical systems via o-minimality of the real numbers (doi)
T. Karimov
ICALP 2025

On the decidability of Presburger arithmetic expanded with powers (doi)
T. Karimov, F. Luca, J. Nieuwveld, J. Ouaknine, and J. Worrell
SODA 2025

*The monadic theory of toric words (doi)
V. Berthé, T. Karimov, J. Nieuwveld, J. Ouaknine, M. Vahanwala, and J. Worrell
Theoretical Computer Science, Vol. 1025 (2024)

On the decidability of monadic second-order logic with arithmetic predicates (doi)
V. Berthé, T. Karimov, J. Nieuwveld, J. Ouaknine, M. Vahanwala, and J. Worrell
LICS 2024, **Distinguished Paper Award**

*Linear dynamical systems with continuous weight functions (doi)
R. Aghamov, C. Baier, T. Karimov, J. Ouaknine, and J. Piribauer
HSCC 2024, **ACM SIGBED Best Paper Award**

*The power of Positivity (doi)
T. Karimov, E. Kelmendi, J. Nieuwveld, J. Ouaknine, and J. Worrell
LICS 2023

Parameter synthesis for parametric probabilistic dynamical systems and prefix-independent specifications (doi)
C. Baier, F. Funke, S. Jantsch, T. Karimov, E. Lefaucheux, J. Ouaknine, D. Purser, M. A. Whiteland, and J. Worrell
CONCUR 2022

*The pseudo-reachability problem for diagonalisable linear dynamical systems (doi)
J. D'Costa, T. Karimov, R. Majumdar, J. Ouaknine, M. Salamati, and J. Worrell
MFCS 2022

*What's decidable about linear loops? (doi)
T. Karimov, E. Lefaucheux, J. Ouaknine, D. Purser, A. Varonka, M. A. Whiteland, and J. Worrell
Proceedings of the ACM on Programming Languages, Vol. 6 (POPL 2022)

The orbit problem for parametric linear dynamical systems (doi)
C. Baier, F. Funke, S. Jantsch, T. Karimov, E. Lefaucheux, F. Luca, J. Ouaknine, D. Purser, M. A. Whiteland, and J. Worrell
CONCUR 2021

*The pseudo-Skolem problem is decidable (doi)
J. D'Costa, T. Karimov, R. Majumdar, J. Ouaknine, M. Salamati, S. Soudjani, and J. Worrell
MFCS 2021

*Deciding ω-regular properties on linear recurrence sequences (doi)
S. Almagor, T. Karimov, E. Kelmendi, J. Ouaknine, and J. Worrell
Proceedings of the ACM on Programming Languages, Vol. 5 (POPL 2021)

Reachability in dynamical systems with rounding (doi)
C. Baier, F. Funke, S. Jantsch, T. Karimov, E. Lefaucheux, J. Ouaknine, A. Pouly, D. Purser, and M. A. Whiteland
FSTTCS 2020

*On LTL model checking for low-dimensional discrete linear dynamical systems (doi)
T. Karimov, J. Ouaknine, and J. Worrell
MFCS 2020

Festschrift and survey papers (peer-reviewed)

Algorithmic applications of Schanuel's conjecture (doi)
T. Karimov, J. Nieuwveld, J. Ouaknine, M. Vahanwala, and J. Worrell
Principles of Formal Quantitative Analysis (Festschrift for Christel Baier)

Model checking Markov chains as distribution transformers (doi)
R. Aghamov, C. Baier, T. Karimov, J. Nieuwveld, J. Ouaknine, J. Piribauer, and M. Vahanwala
Principles of Verification: Cycling the Probabilistic Landscape, Part II (Festschrift for Joost-Pieter Katoen)

*What's decidable about discrete linear dynamical systems? (doi)
T. Karimov, E. Kelmendi, J. Ouaknine, and J. Worrell
Principles of Systems Design (Festschrift for Thomas A. Henzinger)

Papers under peer review

A trichotomy for monadic theories of linear recurrence sequences (pdf)
T. Karimov, J. Nieuwveld, and J. Ouaknine

Preservation theorems for transducer outputs (arXiv)
V. Berthé, H. Goulet-Ouellet, T. Karimov, D. Perrin, and M. Vahanwala

*Rich sequences and decidability of arithmetic theories (arXiv)
T. Karimov, J. Nieuwveld, and J. Ouaknine

On the decidability of monadic theories of arithmetic predicates (arXiv)
V. Berthé, T. Karimov, J. Nieuwveld, J. Ouaknine, M. Vahanwala, and J. Worrell